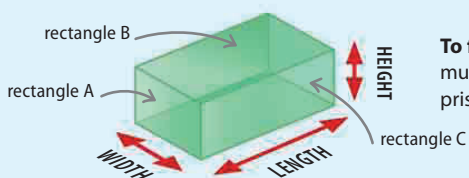


Finding the surface area of a rectangular prism

A rectangular prism is made up of three different pairs of rectangles, here labeled A, B, and C. The surface area is the sum of the areas of all its faces.

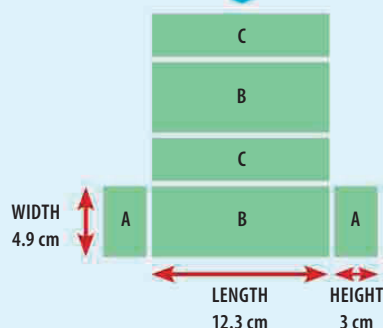


To find the area of rectangle A, multiply together the rectangular prism's height and width.

To find the area of rectangle B, multiply together the rectangular prism's length and width.

To find the area of rectangle C, multiply together the rectangular prism's height and length.

The surface area of the rectangular prism is the total of the areas of its sides—twice area A, added to twice area B, added to twice area C.



△ Net of a rectangular prism

The net is made up of three different pairs of rectangles.

Area of A = height × width

$$3 \times 4.9 = 14.7 \text{ cm}^2$$

Area of B = length × width

$$12.3 \times 4.9 = 60.27 \text{ cm}^2$$

Area of C = height × length

$$3 \times 12.3 = 36.9 \text{ cm}^2$$

parentheses used to separate operations

$$(2 \times A) + (2 \times B) + (2 \times C) \\ (2 \times 14.7) + (2 \times 60.27) + (2 \times 36.9) \\ = 223.74 \text{ cm}^2$$

Finding the surface area of a cone

A cone is made up of two parts—a circular base and a cone shape. Formulas are used to find the areas of the two parts, which are then added together to give the surface area.

Area = $\pi \times r \times h$

slant height
surface area of
cone without base

$$3.14 \times 3.9 \times 9 = 110.21 \text{ cm}^2$$

To find the area of the cone, multiply π by the radius and slant length.

$\pi \times r^2$

formula for area
of a circle
surface area
of base

$$3.14 \times 3.9 \times 3.9 = 47.76 \text{ cm}^2$$

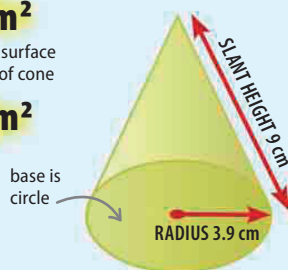
To find the area of the base, use the formula for the area of a circle, $\pi \times r^2$.

$$110.21 + 47.76 = 157.97 \text{ cm}^2$$

total surface
area of cone

▷ Cone

Find the surface area of a cone by using formulas to find the area of the cone shape and the area of the base, and adding the two.



Finding the surface area of a sphere

Unlike many other solid shapes, a sphere cannot be unrolled or unfolded. Instead, a formula is used to find its surface area.

Area = $4 \times \pi \times r^2$

formula for
the surface
area of a
sphere

$$4 \times 3.14 \times 17 \times 17 \\ = 3,629.84 \text{ cm}^2$$

▷ Sphere

The formula for the surface area of a sphere is the same as 4 times the formula for the area of a circle (πr^2). This means that the surface area of a sphere is equal to the surface area of 4 circles with the same radius.

